

Introduction to



with Application to Bioinformatics

- Day 5

Yesterday's Key Takeaways

- Error handling using `try...except`, which helps you catch problems and prevent your program from crashing.
- Keyword arguments and positional arguments, which give you more control, readability, and flexibility when calling functions.
- Docstrings and comments, which provide clear documentation and make your code more understandable and maintainable.
- Pandas, a powerful and widely used library for working with tabular data.

Day 5

- **Session 1**
 - Quiz: Review of Day 4
 - Lecture: Go through questions
 - Lecture: Introduction to regex
 - Ex1: Find the pattern using regex
- **Session 2**
 - Lecture: Regex in Python
 - Ex2: Regex using Python
 - PyQuiz 5.1
- **Session 3**
 - Lecture: Sum up
 - Presentation about NBIS
 - Ex3: Final exercise
- **Project time**

Quiz: Review Day 4

Go to Canvas, Modules -> Day 5 -> Review Day 4

~15 minutes

1. What happens if you declare a variable with the same name inside and outside a function?

The variable inside the function has a separate scope and does not affect the one outside

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The variable inside the function has a separate scope and does not affect the one outside

```
In [ ]: name = "Max"
def changeName():
    name = "Niko"
    print(f"name inside the function: {name}")
changeName()
print(f"name outside of the function: {name}")
```

2. What is the difference between positional arguments and keyword arguments?

Keyword arguments can be given in any order, while positional arguments depend on the function's order

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Keyword arguments can be given in any order, while positional arguments depend on the function's order

```
In [ ]: def sum3(a, b, c):  
        print(f"a={a}")  
        print(f"b={b}")  
        print(f"c={c}")  
        return a+b+c  
sum3(1, 2, 3)
```


2. What is the difference between positional arguments and keyword arguments?

Keyword arguments can be given in any order, while positional arguments depend on the function's order

```
In [ ]: def sum3(a, b, c):  
        print(f"a={a}")  
        print(f"b={b}")  
        print(f"c={c}")  
        return a+b+c  
sum3(1, 2, 3)
```

Arguments can be used in both ways, with or without keyword, if there is no ambiguity

- When used with keyword, they are keyword arguments
- When used without keyword, they are positional arguments

3. What will be the output of the following code snippet?

```
In [ ]: def add(x, y, z=0):  
        return x + y + z  
        print(add(1, 2))  
        print(add(1, y=2, z=3))
```

4. Why is it beneficial to use docstrings in functions?

- They provide explanations and details about the function for others reading your code. Both `"""` and `' '` can be used for docstring

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- They provide explanations and details about the function for others reading your code. Both `"""` and `'''` can be used for docstring

```
In [ ]: def add(x, y, z=0):  
        """This function adds up three numbers, with the last one default to 0  
        # Here we secretly increment x by 1  
        x += 1  
        return x + y + z  
        help(add)
```

5. How can you see the documentation of a Python library function in the console?

- Use `help(library.function)`

```
In [ ]: help(print)
```

6. Which of these import statements would avoid a name conflict if there's a local variable `math` in the same script?

- `import math as m`

```
In [ ]: import math  
math = "great"  
math.sqrt(5)
```

7. What will happen if you import the same module multiple times in a Python script?

- Python ignores subsequent imports of the same module in the same script

If you run

```
import myModule
```

and then update `myModule` and then reload with `import myModule` in Jupyter notebook, the module will not be updated. You will need to run

```
import importlib  
importlib.reload(myModule)
```

8. If you want to filter rows in `df` where `age` is greater than 30, which command would you use?

- `df[df['age'] > 30]`

8. If you want to filter rows in **df** where **age** is greater than 30, which command would you use?

- `df[df['age'] > 30]`

```
In [ ]: import pandas as pd
df = pd.DataFrame({
    'name': ['Alice', 'Bob', 'Charlie', 'David'],
    'age': [25, 30, 35, 40],
    'height': [165.4, 175.3, 168.5, 180.6]
})
print(df)
df[df['age'] > 30]
```

9. If you want to rename multiple columns in a DataFrame `df`, which method should you use?

- `df.rename(columns={'old_col1': 'new_col1', 'old_col2': 'new_col2'})`

If you don't specify the key `columns`, it renames the rows

```
In [ ]: import pandas as pd
df = pd.DataFrame({
    'name': ['Alice', 'Bob', 'Charlie', 'David'],
    'age': [25, 30, 35, 40],
    'height': [165.4, 175.3, 168.5, 180.6]
})
# df.rename(columns={'name': 'SuperStar', 'age': 'TooOld'}, index={0:15,
```

New topic: Regular Expressions

- A regular expression (regex or regexp) is a sequence of characters that defines a search pattern.
- Use case: regular expressions are used in text processing for searching, matching, and manipulating strings

Examples where regex can play a role

- Find variations in a protein or DNA sequence
 - "MVR???A"
 - "ATG???TAG"
- American/British spelling, endings and other variants:
 - salpeter, salpetre, saltpeter, nitre, niter or KNO₃
 - hemaglobin, heamoglobin, hemaglobins, heamoglobin's
 - catalyze, catalyse, catalyzed...
- A pattern in a VCF or GTF file
 - a digit appearing after a tab

Regex is not unique for Python and it is supported by

- most programming languages,
- text editors
- command line tools (e.g. `grep`, `sed`)
- search engines

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- search engines

```
In [ ]: !grep -E "furniture.*sell" ../downloads/blocket_listings.txt
```

A playground for regex with detailed explanations of your regex

<https://regex101.com/>

Defining a search pattern

color
colour
colours
coloring
coloured

} col[ou].r.*

salpeter
salpetre
saltpetre

} salt?pet(er|re)

- `.` matches any character (once)
- `?` repeat previous pattern 0 or 1 times
- `*` repeat previous pattern 0 or more times
- `+` repeat previous pattern 1 or more times

More common operations - classes of characters

- `\w` matches any letter or number, and the underscore
- `\d` matches any digit
- `\D` matches any non-digit
- `\s` matches any whitespace (spaces, tabs, ...)
- `\S` matches any non-whitespace

More common operations - classes of characters

- `\w` matches any letter or number, and the underscore

`\w+`

```
def functionName(arg1, arg2, arg3):  
    final_value = 0  
    # comments  
    return final_value
```

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```

More common operations - classes of characters

- `\d` matches any digit

`\d+`

```
def functionName(arg1, arg2, arg3):  
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    return final_value
```

More common operations - classes of characters

- `\s` matches any whitespace (spaces, tabs, ...)

`\s+`

```
def functionName(arg1, arg2, arg3):  
    final_value = 0  
    # comments  
    return final_value
```

More common operations - classes of characters

- `[abc]` matches a single character defined in this set {a, b, c}
- `[^abc]` matches a single character that is **not** a, b or c
- `[a-z]` matches all letters between `a` and `z` (the english alphabet).
- `[a-z]+` matches any (lowercased) english word.

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- `[a-z]+` matches any (lowercased) english word.

`salt?pet[er]+`

saltpeter

salpetre

~~saltpet88~~

~~salpetin~~

~~saltpet~~

salt?pet[er]+ → salt?pet[er]*

```
saltpeter  
salpetre  
saltpet88  
salpetin  
saltpet
```

`salt?pet[er]+` → `salt?pet[er]*`

```
saltpeter
salpetre
saltpet88
salpetin
saltpet
```

```
In [ ]: # or use https://regex101.com/
! grep -E "salt?pet[er]*" ../tmp/wordlist1.txt
```


Some regular expressions are equivalent; choose the one that you prefer

Shorthand	Equivalent Range	Description
<code>\d</code>	<code>[0-9]</code>	Matches any digit (0-9).
<code>\D</code>	<code>[^0-9]</code>	Matches any non-digit character.
<code>\w</code>	<code>[a-zA-Z0-9_]</code>	Matches any word character (letters, numbers, and the underscore).
<code>\W</code>	<code>[^\w]</code>	Matches any non-word character (e.g., symbols, spaces).
<code>\s</code>	<code>[\t\n\r\f\v]</code>	Matches any whitespace character (space, tab, newline, etc.).
<code>\S</code>	<code>[^\s]</code>	Matches any non-whitespace character.

Example - finding patterns in a VCF file

```
1 920760 rs80259304 T C . PASS
```

```
AA=T;AC=18;AN=120;DP=190;GP=1:930897;BN=131 GT:DP:CB 0/1:1:SM
```

```
0/0:4:SM...
```

If we want to find a record with at least one sample, that is, having genotype fields such as **0/0:1:SM** **0/0:4:SM**

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If we want to find a record with at least one sample, that is, having genotype fields such as **0/0:1:SM** **0/0:4:SM**

- "[01]/[01]" (or "\d/\d")
- \s\d/\d:

Example - finding patterns in a VCF file

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If we want to find a record with at least one sample, that is, having genotype fields such as **0/0:1:SM** **0/0:4:SM**

- "[01]/[01]" (or "\d/\d")
- \s\d/\d:

```
In [ ]: !grep -E "\s\d/\d" ../downloads/genotypes_small.vcf | wc -l
```

Example - finding patterns in vcf

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1 920760 rs80259304 T C . PASS
```

```
AA=T;AC=18;AN=120;DP=190;GP=1:930897;BN=131 GT:DP:CB 0/1:1:SM
```

```
0/0:4/SM...
```

- Find all lines containing more than one homozygous alternate sample (assuming all homozygous alternate are of the form 1/1, which might not be the case in general)

Example - finding patterns in vcf

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0/0:4/SM...
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- Find all lines containing more than one homozygous alternate sample (assuming all homozygous alternate are of the form 1/1, which might not be the case in general)

```
... 1/1:... ... 1/1:... ...
```

Example - finding patterns in vcf

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AA=T;AC=18;AN=120;DP=190;GP=1:930897;BN=131 GT:DP:CB 0/1:1:SM
```

```
0/0:4/SM...
```

- Find all lines containing more than one homozygous alternate sample (assuming all homozygous alternate are of the form 1/1, which might not be the case in general)

```
... 1/1:... .. 1/1:... ..
```

- `.*1/1.*1/1.*`
- `.*\s1/1:.*\s1/1:.*`

```
In [ ]: !grep -E ".*\s1/1:.*\s1/1:.*" ../downloads/genotypes_small.vcf | wc -l
```

Example from the project (with GTF file):

how many transcripts can the CFTR gene (ENSG00000001626) generate?

```
In [ ]: !grep -E "transcript.*ENSG00000001626" ../assignment/data/Homo_sapiens.G
```


Cheat sheet

- `.` matches any character (once)
- `?` repeat previous pattern 0 or 1 times
- `*` repeat previous pattern 0 or more times
- `+` repeat previous pattern 1 or more times
- `\w` matches any letter or number, and the underscore
- `\d` matches any digit
- `\D` matches any non-digit
- `\s` matches any whitespace (spaces, tabs, ...)
- `\S` matches any non-whitespace
- `[abc]` matches a single character defined in this set {a, b, c}
- `[^abc]` matches a single character that is **not** a, b or c
- `[a-z]` matches any (lowercased) letter from the english alphabet
- `.*` matches anything, but **not cross multiple lines**.

Day 5, Exercise 1 (~30 min)

Practicing regular expressions

- Canvas -> Modules -> Day 5 -> Exercise 1 - day 5

Start the exercise by running

python retester.py

in the `downloads` folder in a terminal

Take a break after the exercise (~10 min)

```
%run ../downloads/retester.py
```

Session 2

- How to use regex in Python
- Ex2: Regex using Python
- PyQuiz 5.1

Regular expressions in Python

Regular expressions in Python

```
In [ ]: import re
pattern = r"col[ou]+r.*"
text = "The colour of the wall is very vibrant, but the color of the sky"
# Try to find a hit
result = re.search(pattern, text)
if result:
    print(result.group())
    # print(result.span())
    # print(result.start())
else:
    print("No match found")
```

Regular expressions in Python

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text = "The colour of the wall is very vibrant, but the color of the sky"
# Try to find a hit
result = re.search(pattern, text)
if result:
    print(result.group())
    # print(result.span())
    # print(result.start())
else:
    print("No match found")
```

`result.group()` : Return the string matched by the expression

`result.start()` : Return the starting position of the match

`result.end()` : Return the ending position of the match

`result.span()` : Return both (start, end)

Raw string

```
pattern = r"col[ou]+r.*"
```

- A raw string is a Python string literal prefixed with the letter `r`.
- Highly recommended to use raw string (`r"..."`) for regular expression patterns in Python.
- Raw strings simplify pattern writing by preventing Python from interpreting backslashes `\` as special escape sequences.

Raw string

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- A raw string is a Python string literal prefixed with the letter `r`.
- Highly recommended to use raw string (`r"..."`) for regular expression patterns in Python.
- Raw strings simplify pattern writing by preventing Python from interpreting backslashes `\` as special escape sequences.

```
In [ ]: text = "The colour of the wall is very vibrant, but the color of the sky  
pattern = "\bthe\b" # \b matches the word boundary, try with and without  
result = re.search(pattern, text)  
if result:  
    print(result.group())  
else:  
    print("No match found")
```



```
In [ ]: # How if we only want to search for the words "colour", "color"?  
# pattern = "col[ou]+r.*"  
text = "The colour of the wall is very vibrant, but the color of the sky"  
pattern = r"col[ou]+r\b"  
result = re.search(pattern, text)  
if result:  
    print(f"Found text: '{result.group()}'")  
    print(result.span())  
else:  
    print("No match found")
```

How to find all occurrences of "color" variations?

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```
re.findall
```

How to find all occurrences of "color" variations?

```
re.findall
```

```
In [ ]: text = "The colour of the wall is very vibrant, but the color of the sky  
pattern = r"col[ou]+r\b"  
result = re.findall(pattern, text)  
print(result)
```

How to find all occurrences of "color" variations with their positions in the text?

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```
re.finditer
```

How to find all occurrences of "color" variations with their positions in the text?

```
re.finditer
```

```
In [ ]: text = "The colour of the wall is very vibrant, but the color of the sky  
pattern = r"col[ou]+r\b"  
  
for result in re.finditer(pattern, text):  
    print(f"Find word: {result.group()} at {result.span()}")
```

re.compile

re.compile

```
In [1]: text = "The colour of the wall is very vibrant, but the color of the sky  
pattern = r"col[ou]+r\b"  
p_find_colour = re.compile(pattern)  
p_find_colour.search(text)  
# re.search(pattern, text)
```

re.compile

```
In [ ]: text = "The colour of the wall is very vibrant, but the color of the sky  
pattern = r"col[ou]+r\b"  
p_find_colour = re.compile(pattern)  
p_find_colour.search(text)  
# re.search(pattern, text)
```

```
In [ ]: p_find_colour.findall(text)
```

re.compile

```
In [ ]: text = "The colour of the wall is very vibrant, but the color of the sky  
pattern = r"col[ou]+r\b"  
p_find_colour = re.compile(pattern)  
p_find_colour.search(text)  
# re.search(pattern, text)
```

```
In [ ]: p_find_colour.findall(text)
```

```
In [ ]: for result in p_find_colour.finditer(text):  
        print(f"Find colour at the position {result.span()}, the word is '{r
```

Benefits of using `re.compile`

- Improved performance - saves time by compiling the regex only once
- Reusability - the compiled regex can be used multiple times
- Early error detection - syntax error of the pattern can be detected at the compiling stage instead of mixed with the other `re` functions

Case insensitiveness

```
In [1]: # Remember, [a-z]+ matches any lower case English word
p = re.compile(r'[a-z]+')
result = p.search('123 ATGAAA 456')
if result:
    print(result)
else:
    print("No match found")
```

Case insensitiveness

```
In [ ]: # Remember, [a-z]+ matches any lower case English word
p = re.compile(r'[a-z]+')
result = p.search('123 ATGAAA 456')
if result:
    print(result)
else:
    print("No match found")
```

```
In [ ]: p = re.compile(r'[a-z]+', flags=re.IGNORECASE)

result = p.search('123 ATGAAA 456')
if result:
    print(f"Found hit {result.group()} at the position {result.span()}")
else:
    print("No match found")
```

How to find a full stop?

```
In [1]: text = "The first full stop is here: ."  
        result = re.search(r".", text)  
        print(f"Found {result.group()} at position {result.start()}")
```

How to find a full stop?

```
In [ ]: text = "The first full stop is here: ."  
  
result = re.search(r".", text)  
print(f"Found {result.group()} at position {result.start()}")
```

```
In [ ]: # Use the escape character \  
result = re.search(r"\.", text)  
print(f"Found {result.group()} at position {result.start()}")
```


More operations

- `\` escaping a character, use the character as is
- `^` beginning of the string
- `$` end of string
- `|` boolean `or`

`^hello$`

`^hello$`

```
In [1]: p_find_hello = re.compile(r'^hello$')
text = "hello Python"
result = p_find_hello.search(text)
if result:
    print(result)
else:
    print("No match found")
```

`^hello$`

```
In [1]: p_find_hello = re.compile(r'^hello$')
text = "hello Python"
result = p_find_hello.search(text)
if result:
    print(result)
else:
    print("No match found")
```

```
In [1]: text = "hello"
result = p_find_hello.search(text)
if result:
    print(f"Find word '{result.group()}' at position {result.span()}")
else:
    print("No match found")
```

salt?pet(er|re) | nit(er|re) | KN03

```
salt?pet(er|re) | nit(er|re) | KN03
```

```
In [1]: text = "saltepter or salpeter88 or KN03 or niter or nitre, \  
just too many forms of salpeter!"  
  
p_find_salpeter = re.compile(r'salt?pet(er|re)|nit(er|re)|KN03')  
  
for result in p_find_salpeter.finditer(text):  
    print(f"Find word '{result.group()}' at {result.span()}")
```

Substitution

```
In [1]: text = "Do it becuase I say so, not becuase you want!"
```

Substitution

```
In [ ]: text = "Do it  becuae  I say so,    not becuae you want!"
```

```
In [ ]: # Spell the word because correctly
p_fix_because = re.compile(r'becuae')
p_fix_because.sub('because', text)
print(text)
```


Substitution

```
In [ ]: text = "Do it  becuae  I say so,    not becuae you want!"
```

```
In [ ]: # Spell the word because correctly
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Substitution

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In [ ]: text = "Do it  becuae  I say so,    not becuae you want!"
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```
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p_fix_because.sub('because', text)
print(text)
```

```
In [ ]: text = p_fix_because.sub('because', text)
print(text)
```

```
In [ ]: # Remove additional spaces
p_remove_extra_space = re.compile(r'\s+')
p_remove_extra_space.sub(' ', text)
```

Example from Day 2 when lstrip() does not work as expected

```
In [ ]: line = "AB:A:B should not be removed"  
        line.lstrip("AB:") # The chars argument is not a prefix
```

Example from Day 2 when lstrip() does not work as expected

```
In [ ]: line = "AB:A:B should not be removed"  
        line.lstrip("AB:") # The chars argument is not a prefix
```

```
In [ ]: pattern = r"^AB:"  
        newline = re.sub(pattern, "", line)  
        print(newline)
```

Example from Day 2 when lstrip() does not work as expected

```
In [ ]: line = "AB:A:B should not be removed"
        line.lstrip("AB:") # The chars argument is not a prefix
```

```
In [ ]: pattern = r"^AB:"
        newline = re.sub(pattern, "", line)
        print(newline)
```

```
In [ ]: pattern = r"^AB:"
        p_remove_tag = re.compile(pattern)
        newline = p_remove_tag.sub("", line)
        print(newline)
```

Overview

- Construct regular expressions

```
p = re.compile()
```

- Searching

```
p.search(text)
```

- Substitution

```
p.sub(replacement, text)
```

Typical code structure for text matching:

```
pattern = re.compile(r"regex")
result = pattern.search('string goes here')

if result:
    print('Match found: ', result.group())
else:
    print('No match')
```

Summary

- A powerful tool to search and modify text
- There is much more to read in the [docs](#)
- Note: regex comes in different flavours. If you use it outside Python, there might be small variations in the syntax.

Day 5, Exercise 2 (~30 min)

Use regular expressions with Python

- Canvas -> Modules -> Day 5 -> Exercise 2 - day 5

Take a break after the exercise (~10 min)

PyQuiz 5.1 (~10 min)

Lunch

Sum up!

Processing files - looping through the lines

```
with open('myfile.txt', 'r') as fh:  
    for line in fh:  
        do_stuff(line)
```

Store values

```
iterations = 0
information = []

with open('myfile.txt', 'r') as fh:
    for line in fh:
        iterations += 1
        information += do_stuff(line)
```

Values

- Base types:

```
str    "hello"  
int    5  
float  5.2  
bool   True
```

- Collections:

```
list  ["a", "b", "c"]  
dict  {"a": "alligator", "b": "bear", "c": "cat"}  
tuple ("this", "that")  
set   {"drama", "sci-fi"}
```

Assign values

```
iterations = 0  
score      = 5.2  
# variable = literal
```

Compare and membership

```
+, -, *, ... # mathematical  
and, or, not # logical  
==, !=      # (in)equality  
<, >, <=, >= # comparison  
in          # membership
```

```
In [ ]: value = 4  
        nextvalue = 1  
        nextvalue += value  
        print(f"nextvalue: {nextvalue}, value: {value}")
```

```
In [ ]: value = 4
        nextvalue = 1
        nextvalue += value
        print(f"nextvalue: {nextvalue}, value: {value}")
```

```
In [ ]: x = 5
        y = 7
        z = 0

        x > 4 or y == 7 and z > 1
```



```
In [ ]: value = 4
        nextvalue = 1
        nextvalue += value
        print(f"nextvalue: {nextvalue}, value: {value}")
```

```
In [ ]: x = 5
        y = 7
        z = 0

        x > 4 or y == 7 and z > 1
```

```
In [ ]: (x > 4 or y == 7) and z > 1
```

Strings

Works like a list of characters

```
In [ ]: mystr = "one"
```

```
In [ ]: mystr += " two" # string concatenation  
mystr
```

```
In [ ]: len(mystr) # get the length
```

```
In [ ]: "one" in mystr # in operator for substring matching  
# mystr.find("one") # return index if find a hit  
# mystr.index("one") # Identical to .find(), but handles failure differently
```

String is immutable

```
In [1]: mystr = "one"  
        mystr[1] = "W"
```

String is immutable

```
In [ ]: mystr = "one"  
mystr[1] = "W"
```

```
In [ ]: mystr = "one"  
print(mystr)  
mystr = "two"  
print(mystr)
```

String is immutable

```
In [ ]: mystr = "one"  
        mystr[1] = "W"
```

```
In [ ]: mystr = "one"  
        print(mystr)  
        mystr = "two"  
        print(mystr)
```

```
In [ ]: mystr = "one"  
        print(f"mystr = {mystr}, address = {id(mystr)}")  
        mystr = "two"  
        print(f"mystr = {mystr}, address = {id(mystr)}")
```

String manipulation

```
s.strip() # remove trailing white spaces  
s.split() # split the line into columns  
s.upper(), s.lower() # change the case
```

Regular expressions help you find and replace strings

```
p = re.compile(r'A.A.A')
p.search(dna_seq)

p = re.compile(r'T')
rna_seq = p.sub('U', dna_seq)
```

Regular expressions help you find and replace strings

```
p = re.compile(r'A.A.A')
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p = re.compile(r'T')
rna_seq = p.sub('U', dna_seq)
```

```
In [ ]: import re
p = re.compile(r'p.*\sp') # the greedy star (*)! Match as long as possi
p.search('a python programmer writes python code').group()
```


Regular expressions help you find and replace strings

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p = re.compile(r'A.A.A')
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rna_seq = p.sub('U', dna_seq)
```

```
In [ ]: import re
p = re.compile(r'p.*\sp') # the greedy star (*)! Match as long as possible
p.search('a python programmer writes python code').group()
```

```
In [ ]: # To match the shortest possible string, use non-greedy quantifier *?
p_lazy = re.compile(r'p.*?\sp')
p_lazy.search('a python programmer writes python code').group()
```

Collections

Can contain strings, integer, booleans...

- Most collections are mutable (except **tuple**): you can *add, remove, change* values
- Lists:

```
mylist.append('value')
```

- Dicts:

```
mydict['key'] = 'value'
```

- Sets:

```
myset.add('value')
```

Collections

- Test for membership:

```
value in myobj
```

- Check size:

```
len(myobj)
```

Lists

- Ordered!

```
mylist = [1, 8, 1, 2, 4]
```

```
mylist.sort()
```

```
mylist.reverse()
```

```
mylist[2]
```

```
mylist[-1]
```

```
mylist[2:6]
```

```
In [ ]: mylist = [1, 8, 1, 2, 4]
```

```
In [ ]: mylist.sort()  
print(mylist)
```

```
In [ ]: mylist.reverse()  
print(mylist)
```

```
In [ ]: mylist[2]
```

```
In [ ]: mylist[-1]
```

```
In [ ]: mylist[2:]
```

List multiplication by an integer replicates the list elements the specified number of times.

Question from Day 2

```
In [ ]: mylist * 2
```

List multiplication by an integer replicates the list elements the specified number of times.

Question from Day 2

```
In [ ]: mylist * 2
```

```
In [ ]: import numpy as np
myarray = np.array(mylist)
print(myarray)
print(myarray*2)
```

Dictionaries

- List of key value pairs

```
mydict = {"a": "alligator", "b": "bear", "c": "cat"}  
counter = {"cats": 55, "dogs": 8}
```

```
mydict["a"]  
mydict.keys()  
mydict.values()  
mydict.items() # get both keys and values
```



```
In [ ]: counter = {'cats': 0, 'others': 0}

for animal in ['zebra', 'cat', 'dog', 'cat']:
    if animal == 'cat':
        counter['cats'] += 1
    else:
        counter['others'] += 1

print(counter)
```

Sets

- Bag of values
 - No order
 - No duplicates
 - Fast membership checks
 - Logical set operations (union, difference, intersection...)

```
myset = {"drama", "sci-fi"}
```

```
myset.add("comedy")
```

```
myset.remove("drama")
```

```
In [ ]: set1 = set(["1", "2", "3", "4", "5"])
        set1
```

```
In [ ]: set1 = set(["1", "2", "3", "4", "5"])  
        set1
```

```
In [ ]: set1.add("1")
```

```
In [ ]: set1 = set(["1", "2", "3", "4", "5"])  
set1
```

```
In [ ]: set1.add("1")
```

```
In [ ]: set2 = set(["3", "6"])  
set1.intersection(set2)
```

```
In [ ]: set1 = set(["1", "2", "3", "4", "5"])  
set1
```

```
In [ ]: set1.add("1")
```

```
In [ ]: set2 = set(["3", "6"])  
set1.intersection(set2)
```

```
In [ ]: set1.union(set2)
```

```
In [ ]: set1 = set(["1", "2", "3", "4", "5"])  
set1
```

```
In [ ]: set1.add("1")
```

```
In [ ]: set2 = set(["3", "6"])  
set1.intersection(set2)
```

```
In [ ]: set1.union(set2)
```

```
In [ ]: set1.difference(set2)
```

Tuples

- A group (usually two) of values that belong together

```
tup = (max_length, sequence)
```

- An ordered sequence (like lists)

```
length = tup[0] # get content at index 0
```

- Immutable

Tuples

- A group (usually two) of values that belong together

```
tup = (max_length, sequence)
```

- An ordered sequence (like lists)

```
length = tup[0] # get content at index 0
```

- Immutable

```
In [1]: tup = (2, 'xy')  
tup[0]
```

```
In [ ]: tup[0] = 2
```

Tuples in functions

```
def find_longest_seq(file):  
    # some code here...  
    return length, sequence
```

Tuples in functions

```
def find_longest_seq(file):  
    # some code here...  
    return length, sequence  
  
answer = find_longest_seq(filepath)  
print('length', answer[0])  
print('sequence', answer[1])
```

Tuples in functions

```
def find_longest_seq(file):  
    # some code here...  
    return length, sequence
```

```
answer = find_longest_seq(filepath)  
print('length', answer[0])  
print('sequence', answer[1])
```

```
answer = find_longest_seq(filepath) # return as a tuple  
length, sequence = find_longest_seq(filepath) # return as two  
variables
```

Deciding what to do with if else statement

```
if count > 10:  
    print('big')  
elif count > 5:  
    print('medium')  
else:  
    print('small')
```

```
In [ ]: shopping_list = ['bread', 'egg', ' butter', 'milk']
        tired         = True

        if len(shopping_list) > 4:
            print('Really need to go shopping!')
        elif not tired:
            print('Not tired? Then go shopping!')
        else:
            print('Better to stay at home')
```

Deciding what to do - if statement

Anything that evaluates to a Boolean

```
if condition:  
    print('Condition evaluated to True')  
else:  
    print('Condition evaluated to False')
```

Indentation

if x: is equivalent to **if bool(x):**

Program flow - for loops

```
information = []  
with open('myfile.txt', 'r') as fh:  
    for line in fh:  
        if is_comment(line):  
            use_comment(line)  
        else:  
            information.append(read_data(line)) # read_data return a  
list
```


Program flow - while loops

```
information = []
with open('myfile.txt', 'r') as fh:
    # Read the first line
    line = fh.readline()

    # Continue to read lines until an empty string is returned
    while line:
        information.append(read_data(line)) # read_data return a list
        line = fh.readline() # Read the next line
```

Different types of loops

For loop

is a control flow statement that performs operations over a known amount of steps.

While loop

is a control flow statement that allows code to be executed repeatedly based on a given Boolean condition.

Which one to use?

For loops - standard for iterations over lists and other iterable objects

While loops - more flexible and can iterate an unspecified number of times

```
In [ ]: # For loop example  
# You know the number of iterations before hand  
user_input = "thank god it's friday"  
for letter in user_input:  
    print(letter.upper())
```

```
In [ ]: # For loop example
# You know the number of iterations before hand
user_input = "thank god it's friday"
for letter in user_input:
    print(letter.upper())
```

```
In [ ]: # While loop example
# The number of iterations is unknown before hand
key = ""
while key != "python2025":
    key = input("Enter your key: ")

print("Correct key!")
```

Controlling loops

- `break` - stop the loop
- `continue` - go on to the next iteration

```
In [ ]: # example for break
user_input = "thank god it's friday"
for letter in user_input:
    if letter == 'd':
        break
    print(letter.upper())
```

```
In [ ]: # example for break
user_input = "thank god it's friday"
for letter in user_input:
    if letter == 'd':
        break
    print(letter.upper())
```

```
In [ ]: # example for continue
user_input = "thank god it's friday"
for letter in user_input:
    if letter == ' ' or letter == '\': # Skip spaces and apostrophes
        continue
    print(letter.upper())
```

Watch out!

```
In [ ]: # DON'T RUN THIS  
i = 0  
-while i < 10:  
    print(user_input[i])
```


Watch out!

```
In [ ]: # DON'T RUN THIS  
i = 0  
-while i < 10:  
    print(user_input[i])
```

While loops may be infinite!

File Input/Output

- In: Read from files

```
with open(filename, 'r') as fh:  
    for line in fh:  
        do_stuff(line)
```

Read information from command line: `sys.argv[1:]`

- Out: Write to files:

```
with open(filename, 'w') as fh:  
    fh.write(text)
```

- Printing:

```
print('my_information')
```

Input/Output

Open files should be closed:

```
fh.close()
```

or use the `with` clause

```
with open(filename, "r") as fh:  
    do_something
```

Code structure

- Functions
- Modules

Functions

- A named piece of code that performs a certain task.

```
def functionName(arg1, arg2, arg3):  
    finalValue = 0  
    # Here is some code where you can do  
    # calculations etc, on arg1, arg2, arg3  
    # and update finalValue  
    return finalValue
```

- Is given a number of input arguments
 - to be used within the function body
- Returns a result (maybe `None`)

Functions - arguments

```
def prettyprinter(name, value, delim=":", end=None):  
    out = "The " + name + " is " + delim + " " + value  
    if end:  
        out += end  
    return out
```

- used to set default values (often `None`)
- can be skipped in function calls, then the arguments are positional
- improve readability when keys are used

Using your code

Any longer pieces of code that have been used and will be re-used should be saved

- Save it as a file `mycode.py`
- To run it: `python3 mycode.py` or `python mycode.py`
- Import it: `import mycode`

Documentation and comments

```
""" This is a doc-string explaining what the purpose of this function/  
module is """
```

```
# This is a comment that helps understanding the code
```


Documentation and comments

```
""" This is a doc-string explaining what the purpose of this function/  
module is """
```

```
# This is a comment that helps understanding the code
```

- Comments *will* help you
- Undocumented code rarely gets used
- Try to keep your code readable: use informative variable and function names

Why programming?

Endless possibilities!

- reverse complement DNA
- custom filtering of VCF files
- plotting of results
- dealing with excel files

Why programming?

- Computers are fast
- Computers don't get bored
- Computers don't get sloppy

Why programming?

- Computers are fast
- Computers don't get bored
- Computers don't get sloppy
- Create reproducible results
 - for you and for others to use
- Extract large amount of information

Type hinting: popular for modern Python coding

```
In [ ]: age: int = 30
        name: str = "Bob"

        def show_new_age(age: int, name: str) -> int:
            # 'age' is expected to be an integer
            # 'name' is expected to be a string
            # The function is expected to return an integer
            new_age = age + 10
            print(f"New age of {name} is {new_age}")
            return new_age
        show_new_age(age, name)
```

Type hinting: popular for modern Python coding

```
In [ ]: age: int = 30
        name: str = "Bob"

def show_new_age(age: int, name: str) -> int:
    # 'age' is expected to be an integer
    # 'name' is expected to be a string
    # The function is expected to return an integer
    new_age = age + 10
    print(f"New age of {name} is {new_age}")
    return new_age
show_new_age(age, name)
```

- Type hinting improve the documentation and readability
- With proper tools and IDE, type hinting helps early detection of errors and debugging

```
In [ ]: age: int = 30.0  
name: str = "Bob"  
print(type(age))
```

```
In [ ]: age: int = 30.0  
        name: str = "Bob"  
        print(type(age))
```

Type hinting does not change how Python interprets your code

Final advice

- Take a moment to think before you start coding
 - use pseudocode
 - use top-down programming
 - use paper and pen
 - take breaks

Final advice

- Take a moment to think before you start coding
 - use pseudocode
 - use top-down programming
 - use paper and pen
 - take breaks
- You know the basics - don't be afraid to try, it's the only way to learn
- You will get faster

Final advice (for real)

- Getting help
 - ask colleagues
 - try talk about your problem (get a rubber duck https://en.wikipedia.org/wiki/Rubber_duck_debugging)
 - search the web
 - Ask AI (such as chatGPT)
 - [NBIS drop-ins](#)

Now you know Python!



Well done!

NBIS presentation

Just a small exercise to finish the day and have fun!

Canvas -> Module -> Day 5 -> Exercise 3 - day 5

[Google Form for anonymous evaluation](#)